

Cooperative Membership as Partial Climate Adaptation: Transitory Withdrawal and Permanent Exit

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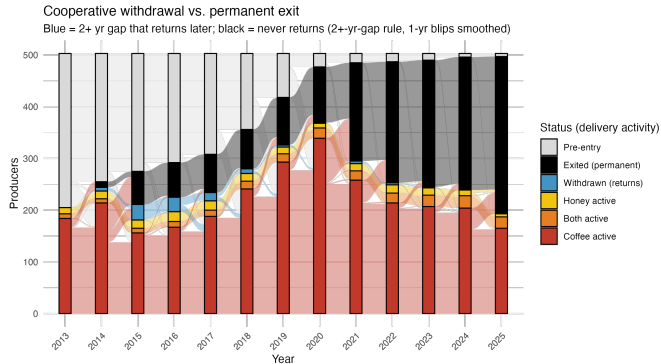
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AAEA Annual Meeting
27 July 2026

Motivation: A Climate Adaptation That Farmers Abandon

- Cooperative membership is a **climate-adaptation technology**, and it adapts **through prices**: a floor price and pooled risk are how the co-op turns a drought or leaf-rust shock into smoothed income.
- Yet in our Chiapas cooperative, **active coffee members fell from 356 (2020) to 180 (2026)** — farmers walking away from the adaptation.
- **But the decline is not monotonic.** Two distinct episodes, with different drivers, sit behind the net fall.
- The puzzle: *when* does cooperative membership deliver adaptation — and what happens when the price channel that delivers it breaks down?

Two Exit Waves: Withdrawal vs. Permanent Exit



- **2015 drought** (coincident with the regional roya outbreak): a temporary *withdrawal* — producers leave, then *return*. Adaptation works.
- **Post-2020:** *permanent* exit, coinciding with a negative **price spread** (co-op floor price — local/coyote price) that pushes members to side-sell (Keenan et al., 2024; Milford, 2014). Adaptation fails.

Cooperative Membership as *Partial* Adaptation

- Adaptation offsets only part of climate damage (Burke & Emerick, 2016) — the cooperative is one **partial** layer, not a full hedge.
- Shocks vary on two axes: **who** they hit — *idiosyncratic* vs. *covariate* — and **how long** they last — *transitory* vs. *permanent*.
- The co-op covers three of the four cells: its **price floor** insures the idiosyncratic price downside (Bellemare et al., 2021); **microcredit and honey** smooth transitory shocks over time. It cannot hedge a shock that is both *covariate and persistent* — a village-wide price collapse that does not reverse.
- **Drought is the covariate axis**: village-wide, but transitory, so members withdraw and return — yet it still blunts the network channel that spreads membership.
- Whether it takes hold depends on how membership **diffuses through village networks** (Beaman et al., 2021; Munshi, 2014).

Motivation: Identification Challenges

- Bramoullé et al. (2020) gives conditions for causally identifying peer effects.
- Individual farmers' decisions to adopt may be affected by:
 - Unobservable characteristics of those farmers.
 - Adoption decisions of neighbors (**endogenous peer effects**).
 - Other characteristics of neighbors (**contextual peer effects**).
- Distinguishing between endogenous peer effects and contextual peer effects is the “reflection problem” (Manski, 1993).
- In addition, unobserved shocks may affect the adoption decision of both a farmer and their neighbors (**correlated effects**).
- Our solution: geographic exogeneity of networks + drought shocks as exogenous shifters.

This Paper

- 23-year panel of producer entry decisions into a coffee and/or honey cooperative.
- We use two technologies on the same fixed network as mutual placebos.
- We use a linear-in-means model to examine entry over time.
- We interact peer effects with drought shocks.
- **Temporal friction:** compare fixed effects and first differences specifications.
- **Spatial friction:** compare direct and indirect effects using a spatial lag model.
- We interpret asymmetric results in a homophily vs. contagion framework.

Preview of Findings

- **Key finding — partial adaptation:** The co-op absorbs shocks that are *idiosyncratic* or *transitory*, but not one that is both *covariate* and *persistent*.
- **Two waves:** the *transitory* 2015 drought → producers withdraw and then *return* — smoothing works; the *persistent*, village-wide adverse price spread after 2020 → *permanent exit*, sharpest where the co-op most under-pays — smoothing fails.
- **Network channel and its limit:** Membership diffuses through village networks (a 10% higher village entry rate → ~5% higher own entry probability; the effect drops ~75–80% from fixed effects to first differences, i.e. long-run adaptation) — but the covariate drought *counteracts* that channel (negative drought × network interaction).

Contribution

- We use two technologies on the same fixed network as mutual placebos to separate homophily from contagion in technology adoption.
- We show how comparing fixed effects and first differences specifications reveals temporal frictions in information diffusion.
- We show how comparing direct and indirect spatial lag effects reveals spatial frictions in information diffusion.
- We provide a 23-year panel of producer entry decisions from an indigenous community in Chiapas, Mexico.

Study Area



Indigenous Community

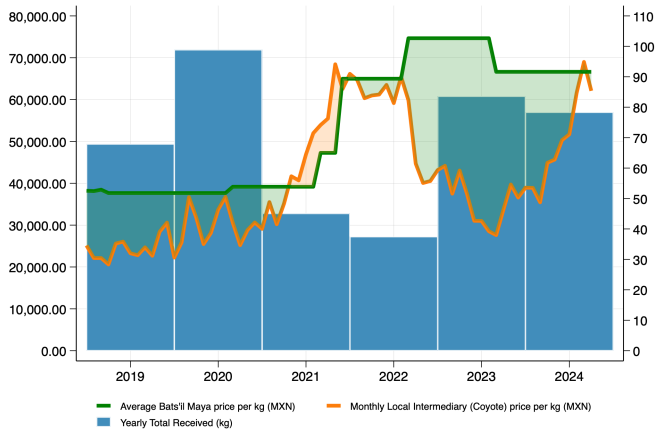
- Tzeltal Maya communities in the highlands of Chiapas, Mexico.
- Producers are organized into villages with stable social networks rooted in geography and culture.
- Coffee and honey production are both non-native technologies introduced during the 20th century.
- Two cooperatives — one for coffee, one for honey — serve the same producer population.



Coffee Cooperative

- **Price volatility** → **price insurance**: The cooperative guarantees a floor price for coffee, insuring producers against global market fluctuations.
- **Coffee rust shock** → **better plants and techniques**: The cooperative provides access to disease-resistant varieties and technical training in response to coffee leaf rust outbreaks.
- **Consumption smoothing** → **emergency microloans**: The cooperative offers small emergency loans to help producers bridge income gaps between harvests.

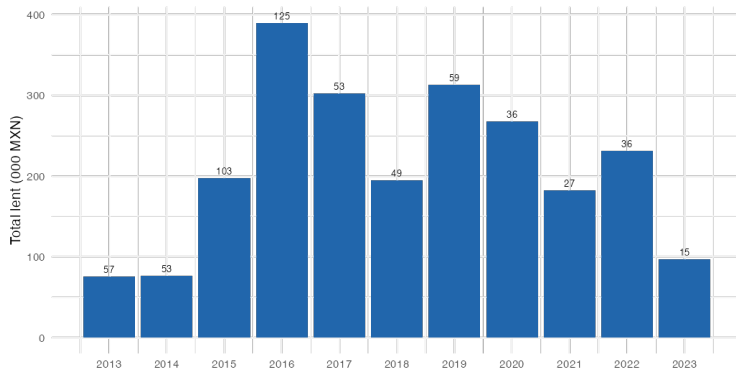
Price Insurance



Cooperative Microcredit

Microcredit volume by year (bar = total pesos '000; label = # loans)

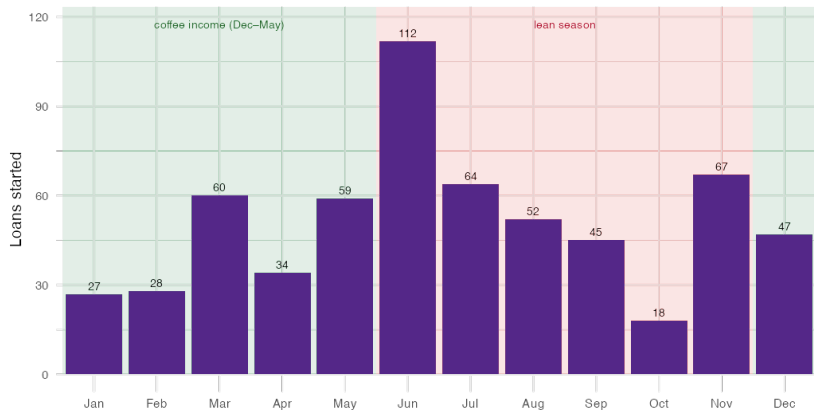
251 borrowers, 626 loans, 2013–2023; median loan 2,000 MXN



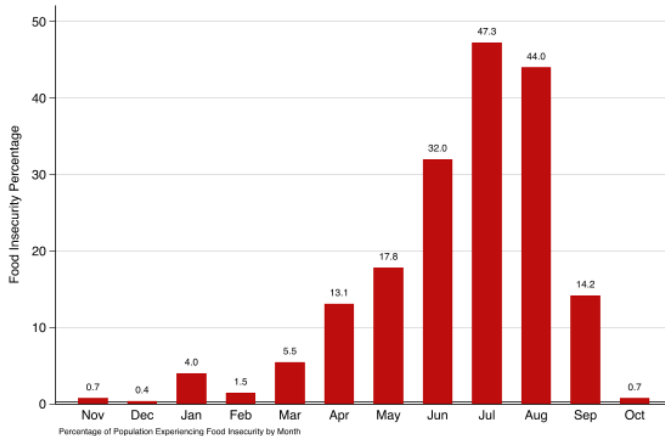
Borrowing Peaks in the Lean Season

Borrowing peaks in the lean season between coffee harvests

Loan starts by calendar month, 2013–2023



Food Insecurity Exposure



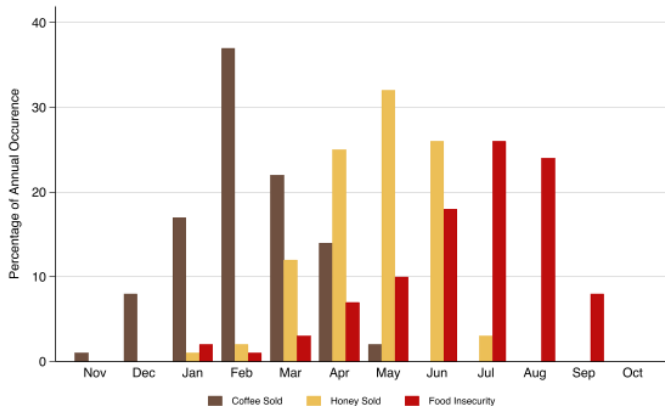
Honey Cooperative

- **Income at another time of year:** Honey harvest arrives at a different point in the agricultural calendar, smoothing consumption across the year.
- **A sweet and timely form of insurance:** turn on and off with small capital and labor investment.
- **Differential food insecurity:** Producers who adopt honey production experience less food insecurity than those who rely on coffee alone.

A Sweet and Timely Form of Insurance

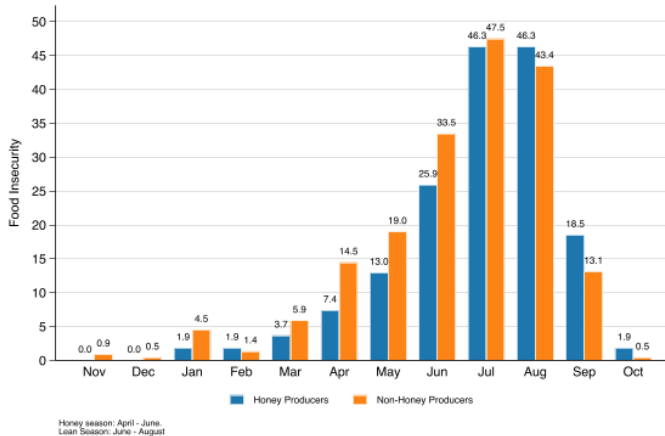


Coffee and Honey Seasonality



Coffee sales source: Yomoi ATel 2013 - 2019 receipts.
Honey sales receipts: Yomoi ATel 2022 receipts.

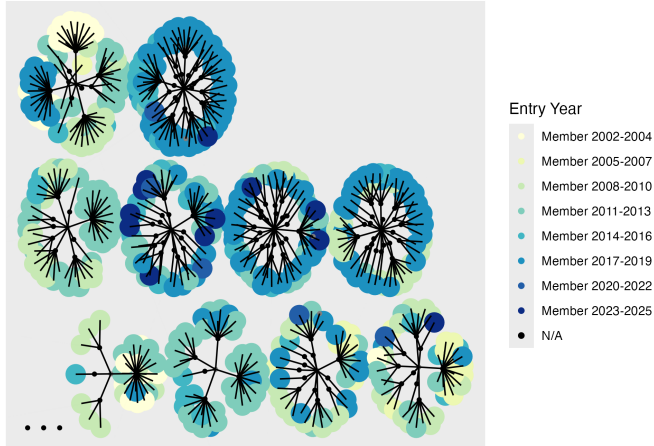
Honey Buffers, but Doesn't Eliminate, the Lean Season



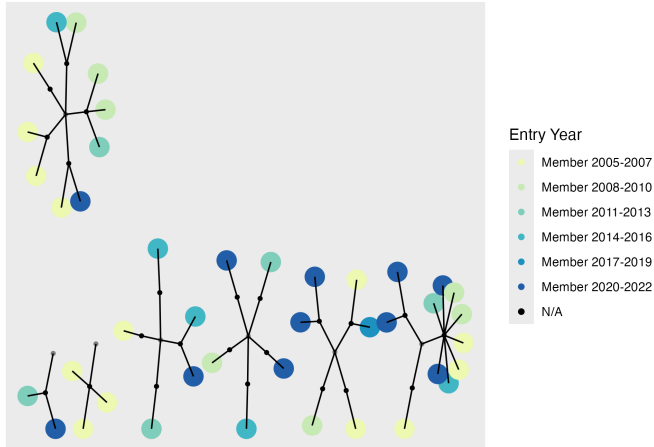
Producers and Villages

- Universe of 498 producers who adopt into the coffee cooperative, the honey cooperative, or both.
- Spatial variation: eleven regions divided into 124 total villages.
- Network definition: producers in the same village share a direct link; producers in the same region but different village share an indirect link.
- Temporal variation: 23 years
 - 2002–2012: self-reported date of entrance
 - 2013–2025: deliveries to both cooperatives

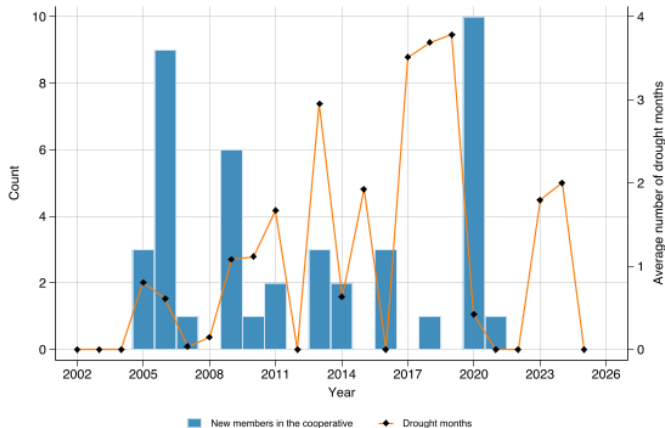
Network Structure of Coffee Cooperative Entry



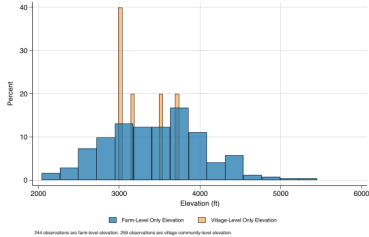
Network Structure of Honey Cooperative Entry



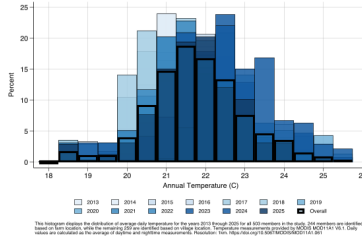
Honey Cooperative Entry and Drought



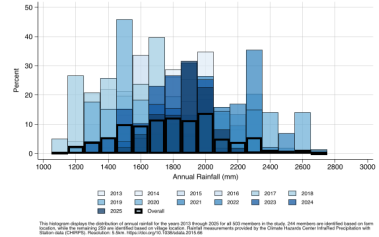
Climate Characteristics



Elevation

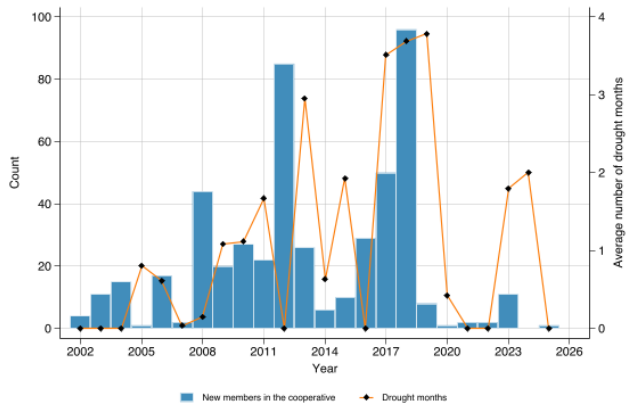


Temperature



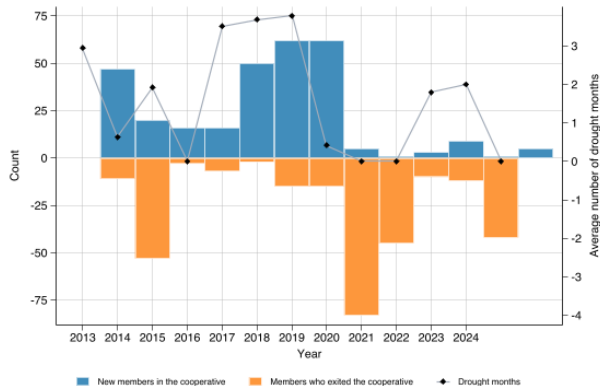
Rainfall

Drought Partially Explains Entry...

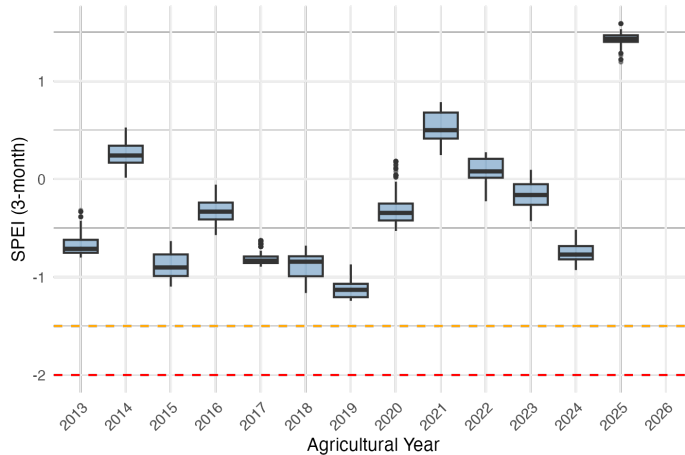


... But Not Exit

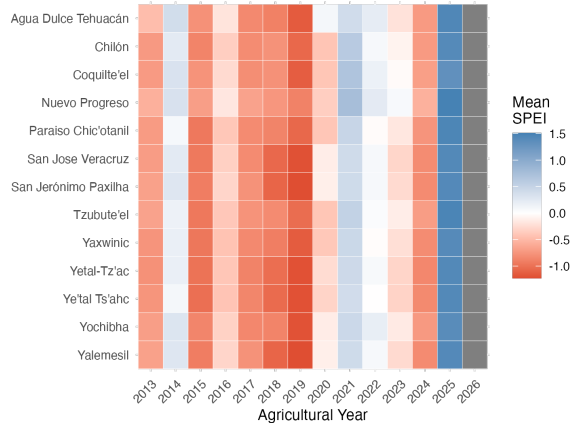
Exit is driven by the coffee market — each producer's idiosyncratic co-op—local price spread — not by drought.



SPEI Distribution by Year



Mean SPEI by Region and Year



Empirical Strategy - Linear-in-Means Model

- We use a linear-in-means model to examine the peer effects of producers in the same village on adoption (Bramoullé et al., 2009).
- The outcome is y_{ijt}^z : whether producer i in village j adopts cooperative z in time t .
- N_{ijt}^z in Equation (1) is network strength; D_{ijt} is number of periods of drought.
- Following Millimet and Bellemare (2025), we augment our producer fixed-effect specification in Equation (2) with a first-difference one in Equation (3).

$$N_{ijt}^z = \frac{1}{n_j - 1} \sum_{k=1, k \neq i}^{n_j} y_{kjt}^z \quad (1)$$

$$y_{ijt}^z = \alpha_1^z + \beta_1^z N_{ijt-1}^z + \delta_1^z \mathbf{D}_{ijt-1} + \gamma_1^z \mathbf{D}_{ijt-1} N_{ijt-1}^z + \phi_{1i}^z + \xi_{1t}^z + \epsilon_{1ijt}^z \quad (2)$$

$$\Delta y_{ijt}^z = \alpha_2^z + \beta_2^z \Delta N_{ijt-1}^z + \delta_2^z \Delta \mathbf{D}_{ijt-1} + \gamma_2^z \Delta \mathbf{D}_{ijt-1} N_{ijt-1}^z + \xi_{2t}^z + \Delta \epsilon_{2ijt}^z \quad (3)$$

Empirical Strategy - Spatial Lag Model

- We use a spatial lag model to examine additional indirect effects of adoption decisions in other villages in the same region or all other villages in addition to the direct effects of peer effects in a producer's own village (Halleck Vega & Elhorst, 2015).

$$y_{ijt}^z = \alpha_3^z + \beta_3^z N_{ijt-1}^z + \delta_3^z \mathbf{D}_{ijt-1} + \mathbf{W} \mathbf{N}_{t-1}^z \theta_3 + \mathbf{W} \mathbf{D}_{t-1} \lambda_3 + \epsilon_{3ijt}^z \quad (4)$$

- Network strength N and drought D are defined as above.
- We build three weighting matrices $\mathbf{W}$. Diagonal elements are 0. Off diagonal elements are as follows for producer j and k .
 - ① **Binary Contiguity:** $w_{jk}^1 = 1$ iff j and k are in the same region; 0 otherwise.
 - ② **Inverse Distance - Region:** $w_{jk}^2 = \frac{1}{d_{jk}}$ iff j and k are in the same region; 0 otherwise.
 - ③ **Inverse Distance - All:** $w_{jk}^3 = \frac{1}{d_{jk}}$

Identification

- Recall that network structure is stable over the long term.
- Geographic networks are plausibly exogenous — little mobility means producers did not sort into networks based on cooperative membership.
- Drought shocks provide identifying variation: conditional on network structure, drought is an exogenous shifter of adoption incentives.
- **Temporal frictions:** A large drop in the peer effect from FE to FD implies the effect is driven by long-run adaptation rather than recent adoption.
- **Spatial frictions:** Stronger indirect than direct effects imply that information travels beyond the village in adoption decisions.

Coffee Cooperative - Linear-in-Means

Network effect drops from ~50% (FE) to ~10% (FD) — look at columns 1 vs. 3.

	Joins Coffee Cooperative (1=Yes)			
	FE Short	FE Long	FD Short	FD Long
	(1)	(2)	(3)	(4)
Network Strength (Village)	0.539*** (0.038)	0.522*** (0.029)	0.102*** (0.028)	0.004 (0.014)
Periods of Severe Drought	0.046*** (0.008)	0.024*** (0.006)	0.026*** (0.007)	0.005 (0.004)
Periods of Extreme Drought	-0.050** (0.024)	-0.019 (0.014)	0.002 (0.011)	-0.019** (0.008)
Severe Drought x Network Strength	-0.042*** (0.007)	-0.039*** (0.006)	-0.037*** (0.006)	-0.018*** (0.005)
Extreme Drought x Network Strength	0.049*** (0.017)	-0.028** (0.013)	-0.008 (0.010)	0.009 (0.006)
Year Fixed Effects	Y	Y	Y	Y
Producers	503	503	503	503
Years	12	23	12	23
Observations	6,036	11,569	5,533	11,066
Adjusted R ²	0.085	0.116	0.098	0.065

Honey Cooperative - Linear-in-Means

Same network effect drop, but drought interactions are insignificant throughout.

	Joins Honey Cooperative (1=Yes)			
	FE Short	FE Long	FD Short	FD Long
	(1)	(2)	(3)	(4)
Network Strength (Village)	0.519*** (0.124)	0.441*** (0.104)	0.126 (0.123)	0.072 (0.071)
Periods of Severe Drought	-0.0003 (0.001)	-0.001 (0.002)	-0.0003 (0.001)	0.001 (0.001)
Periods of Extreme Drought	0.002 (0.005)	-0.002 (0.004)	0.001 (0.003)	0.004* (0.002)
Severe Drought × Network Strength	-0.006 (0.008)	-0.031*** (0.010)	0.005 (0.005)	0.001 (0.005)
Extreme Drought × Network Strength	0.001 (0.011)	0.030** (0.014)	0.001 (0.002)	-0.002 (0.002)
Year Fixed Effects	Y	Y	Y	Y
Producers	503	503	503	503
Years	12	23	12	23
Observations	6,036	11,569	5,533	11,066
Adjusted R ²	0.049	0.027	0.018	0.008

LiM Results

- For coffee: severe drought $\rightarrow$ adoption $\uparrow$, but network effect offsets this; extreme drought $\rightarrow$ adoption $\downarrow$.
- For coffee: the FE to FD drop ($\sim 50\%$ to $\sim 10\%$) suggests long-run adaptation, not recent peer influence.
- For honey: over the long term, extreme drought in high-network villages increases adoption; severe drought decreases it.
- For honey: the FE to FD drop is similar to coffee ($\sim 50\%$ to $\sim 13\%$), confirming the network effect reflects long-run adaptation — but drought interactions are absent, unlike coffee.

Spatial Lag: How Far Adaptation Travels

- **Coffee:** indirect (cross-village) network and drought effects *exceed* the direct within-village effects — information about the cooperative and about climate risk diffuses **across** villages (direct network 30–40%; indirect ~40–75%).
- **Honey: direct network effects only** (40–51%) — no significant cross-village spillover and no drought interaction; a newer technology with less codified climate-resilience information.
- **Why partial:** the network transmits adaptation spatially for coffee but stalls locally for honey — diffusion, and hence adaptation, is incomplete.
- *Robustness:* the negative indirect honey effect is FE-dependent (does not survive dropping year FE); direct network and coffee spillover channels are robust.

Backup tables:

▶ Coffee SLX

▶ Honey SLX

One Interpretation: Separating Homophily from Contagion

Challenge: Network correlations in adoption could reflect homophily (similar producers cluster) rather than genuine peer effects. We are inspired by Golub and Jackson (2012).

Our strategy: Use two technologies on the same fixed network as mutual placebos.

Under pure homophily

- Fixed producer characteristics ϕ_{1i}^Z drive clustering
- Network effects should be proportional across technologies
- Drought interactions should not differ

Under contagion

- β_1^Z reflects technology-specific information value
- Drought interactions reflect information about *climate resilience*
- Coffee and honey effects can diverge

Adaptation Is Partial

What we find:

- Cooperative membership diffuses through networks and adapts producers — but a covariate drought shock *counteracts* the network channel (negative drought $\times$ network).
- **Same network, opposite responses:** the drought $\times$ network interaction is strong for coffee but *absent* for honey — so the asymmetry isn't a network artifact; drought disrupts the coffee-adaptation channel in particular.

Interpretation — adaptation is partial:

- The co-op smooths *idiosyncratic* and *transitory* shocks (price floor; microcredit and honey over time), but not a *permanent, covariate* drought that hits the whole network at once.

Next steps

- Ground *partial adaptation* in risk-sharing theory — idiosyncratic vs. covariate insurance (Fafchamps & Lund, 2003; Mobarak & Rosenzweig, 2013; Townsend, Robert, 1994).
- Incorporate the **microcredit** data (the cooperative's emergency-loan channel) into the analysis — a further consumption-smoothing / insurance layer.
- Analyze the price spread more closely, linking to Pitts et al. (2025).

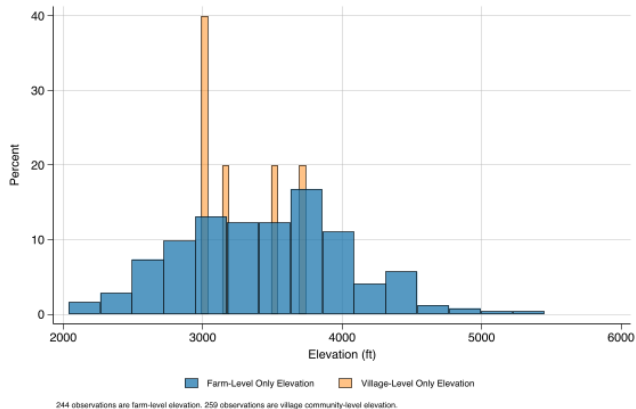
Thank you!

Appendix

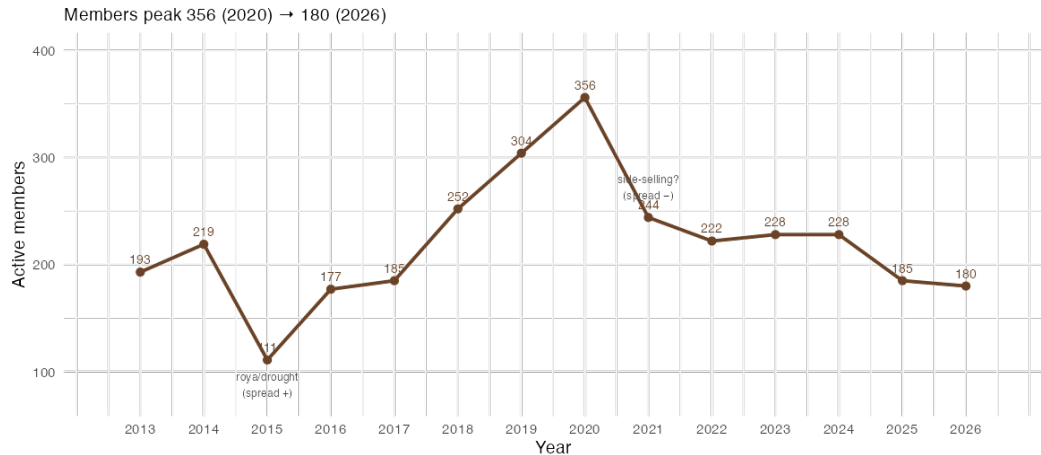
This Paper (original slide 4)

- We leverage a unique 23 year panel of entry decisions by indigenous producers.
- Producers enter a coffee cooperative, a honey cooperative, or both.
- Adoption is an absorbing state (for now).
- Network structure exhibits the "small worlds" property (Jackson, 2010).
- We argue network structure is stable for cultural reasons.
- We use drought shocks to identify peer effects.
- Conditional on experiencing a drought, we estimate the effect of last year's village adoption rate on this year's probability of adoption.

Elevation Distribution



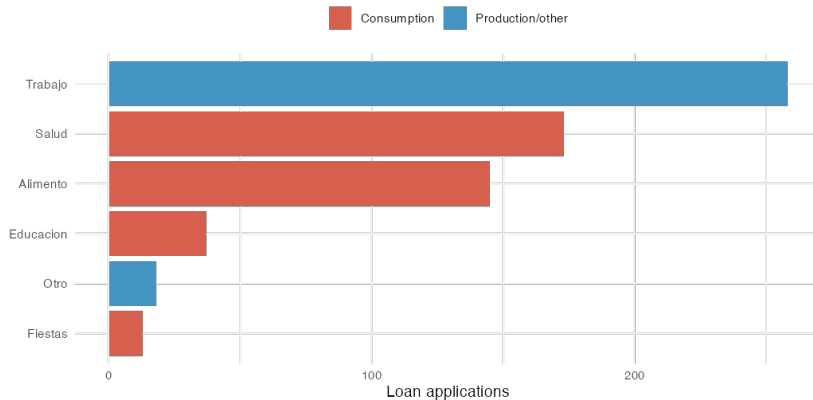
Active Coffee Membership, 2013–2026 (backup)



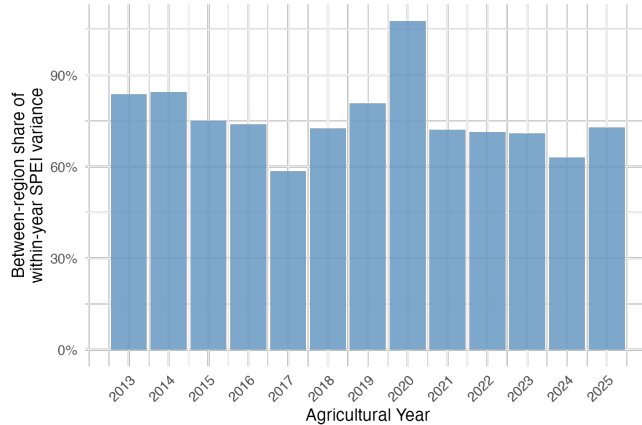
Loan Purpose Is Mostly Consumption (backup)

Loan purpose is mostly consumption smoothing

Health + food + education + fiestas (red) > work/production (blue)



Within-Year ICC of SPEI by Region



Adoption by Village

			Coffee			Honey		
	Region	Villages	2002	2013	2025	2005	2013	2025
	1	5	0	4	5	0	0	0
	2	15	0	7	15	1	2	4
	3	9	0	9	9	0	1	1
	4	9	1	7	8	0	5	5
	5	4	0	3	4	1	1	1
	6	21	0	9	21	1	3	4
	7	9	0	7	8	0	2	3
	8	20	0	8	19	0	2	5
	9	19	0	1	19	0	0	0
	10	12	0	8	12	0	0	0
Total	—	123	1	63	120	3	16	23

This table shows the number of villages in each region with members of the coffee and honey cooperatives at three time periods: the first year, a middle year, and the latest year in the dataset.

Adoption by Individual

			Coffee			Honey		
	Region	Individuals	2002	2013	2025	2005	2013	2025
	1	39	0	29	39	0	0	0
	2	51	0	18	48	1	2	5
	3	52	0	49	52	0	1	2
	4	72	4	53	67	0	8	10
	5	28	0	26	28	1	3	3
	6	52	0	18	52	1	3	6
	7	43	0	34	42	0	6	11
	8	58	0	21	54	0	2	5
	9	72	0	1	72	0	0	0
	10	36	0	25	36	0	0	0
Total	—	503	4	274	490	3	25	42

This table shows the number of individuals in each region with members of the coffee and honey cooperatives at three time periods: the first year, a middle year, and the latest year in the dataset.

Coffee Cooperative — SLX (backup)

Indirect network effects exceed direct effects — and indirect drought effects exceed direct.

	Joins Coffee Cooperative (1=Yes)					
	(1)	(2)	(3)	(4)	(5)	(6)
Direct Effect - Network	0.303*** (0.023)	0.339*** (0.022)	0.386*** (0.022)	0.347*** (0.014)	0.350*** (0.013)	0.387*** (0.013)
Direct Effect - Severe Drought	0.006 (0.009)	0.012 (0.009)	0.0002 (0.009)	-0.006 (0.007)	-0.005 (0.008)	-0.016** (0.007)
Direct Effect - Extreme Drought	-0.017 (0.021)	0.002 (0.023)	-0.027 (0.020)	-0.021 (0.019)	-0.016 (0.019)	-0.041** (0.017)
Indirect Effect - Network	0.592*** (0.033)	0.472*** (0.031)	0.777*** (0.057)	0.452*** (0.022)	0.431*** (0.018)	0.813*** (0.039)
Indirect Effect - Severe Drought	0.021** (0.010)	0.017* (0.010)	0.082*** (0.018)	0.027*** (0.009)	0.021*** (0.008)	0.072*** (0.015)
Indirect Effect - Extreme Drought	0.047* (0.025)	0.019 (0.026)	0.197*** (0.051)	0.012 (0.021)	-0.001 (0.021)	0.079** (0.039)
Weighting Formula	Binary	Inv Dist	Inv Dist	Binary	Inv Dist	Inv Dist
Other Villages	Region Only	Region Only	All	Region Only	Region Only	All
Years	12	12	12	23	23	23
Log-Likelihood	831	787	735	-152	-72	-169
Observations	6,036	6,036	6,036	11,569	11,569	11,569

Honey Cooperative — SLX (backup)

Direct network effects only — no significant indirect effects or drought interactions.

	Joins Honey Cooperative (1=Yes)					
	(1)	(2)	(3)	(4)	(5)	(6)
Direct Effect - Network	0.513*** (0.059)	0.516*** (0.059)	0.515*** (0.059)	0.415*** (0.035)	0.415*** (0.035)	0.410*** (0.035)
Direct Effect - Severe Drought	-0.002 (0.002)	-0.004 (0.004)	-0.005 (0.006)	-0.003 (0.002)	0.002 (0.003)	0.006 (0.005)
Direct Effect - Extreme Drought	-0.002 (0.006)	-0.006 (0.011)	-0.016 (0.014)	-0.009 (0.006)	0.007 (0.009)	0.011 (0.010)
Indirect Effect - Network	0.090 (0.056)	-0.044* (0.025)	-0.233*** (0.056)	0.080* (0.041)	0.010 (0.032)	0.270*** (0.088)
Indirect Effect - Severe Drought	0.004 (0.003)	0.005 (0.005)	0.019 (0.018)	0.001 (0.003)	-0.006 (0.004)	-0.019 (0.013)
Indirect Effect - Extreme Drought	0.008 (0.006)	0.013 (0.014)	0.065 (0.051)	0.013* (0.007)	-0.012 (0.011)	-0.032 (0.032)
Weighting Formula	Binary	Inv Dist	Inv Dist	Binary	Inv Dist	Inv Dist
Other Villages	Region Only	Region Only	All	Region Only	Region Only	All
Years	12	12	12	23	23	23
Log-Likelihood	7028	7032	7047	8598	8598	8620
Observations	6,036	6,036	6,036	11,569	11,569	11,569

Robustness: LiM Coffee — Drop Year FE

Story intact; coffee magnitudes inflate (year FE is the conservative spec).

	Joins Coffee Cooperative (1=Yes)			
	FE Short (1)	FE Long (2)	FD Short (3)	FD Long (4)
Network Strength (Village)	0.760*** (0.031)	0.914*** (0.011)	0.148*** (0.028)	0.025* (0.014)
Periods of Severe Drought	0.060*** (0.007)	0.087*** (0.005)	0.039*** (0.006)	0.016*** (0.004)
Periods of Extreme Drought	−0.073*** (0.016)	0.017 (0.011)	0.016* (0.009)	−0.009* (0.005)
Severe Drought x Network Strength	−0.059*** (0.008)	−0.083*** (0.006)	−0.039*** (0.007)	−0.012*** (0.004)
Extreme Drought x Network Strength	0.081*** (0.017)	−0.027** (0.013)	−0.002 (0.009)	0.011* (0.006)
Year Fixed Effects	N	N	N	N
Producers	503	503	503	503
Years	12	23	12	23
Observations	6,036	11,569	5,533	11,066
Adjusted R ²	0.307	0.610	0.048	0.006

Robustness: LiM Honey — Drop Year FE

Honey estimates stable when year FE dropped.

	Joins Honey Cooperative (1=Yes)			
	FE Short (1)	FE Long (2)	FD Short (3)	FD Long (4)
Network Strength (Village)	0.536*** (0.121)	0.504*** (0.101)	0.121 (0.123)	0.070 (0.071)
Periods of Severe Drought	0.0003 (0.0005)	0.005*** (0.001)	0.001** (0.0003)	0.0004* (0.0002)
Periods of Extreme Drought	0.001 (0.001)	0.005*** (0.002)	−0.001 (0.001)	0.002 (0.001)
Severe Drought × Network Strength	−0.007 (0.007)	−0.038*** (0.010)	0.004 (0.005)	0.0004 (0.005)
Extreme Drought × Network Strength	0.006 (0.011)	0.023* (0.014)	0.003 (0.002)	−0.004 (0.003)
Year Fixed Effects	N	N	N	N
Producers	503	503	503	503
Years	12	23	12	23
Observations	6,036	11,569	5,533	11,066
Adjusted R ²	0.066	0.061	0.008	0.001

Robustness: LiM Coffee — Continuous SPEI

Continuous SPEI confirms and sharpens the drought story (coffee SPEI significant throughout; SPEI lower = drier).

	Joins Coffee Cooperative (1=Yes)			
	FE Short	FE Long	FD Short	FD Long
	(1)	(2)	(3)	(4)
Network Strength (Village)	0.727*** (0.028)	0.875*** (0.010)	0.095*** (0.023)	0.016 (0.011)
SPEI (Prior Year)	-0.071*** (0.013)	-0.169*** (0.009)	-0.054*** (0.011)	-0.017*** (0.004)
SPEI x Network Strength	0.069*** (0.013)	0.173*** (0.011)	0.053*** (0.011)	0.014** (0.006)
Year Fixed Effects	N	N	N	N
Producers	503	503	503	503
Years	12	23	12	23
Observations	6,036	11,569	5,533	11,066
Adjusted R ²	0.290	0.607	0.010	0.002

Robustness: LiM Honey — Continuous SPEI

Honey remains network-driven; little drought response under continuous SPEI.

	Joins Honey Cooperative (1=Yes)			
	FE Short	FE Long	FD Short	FD Long
	(1)	(2)	(3)	(4)
Network Strength (Village)	0.536*** (0.121)	0.492*** (0.099)	0.122 (0.122)	0.067 (0.070)
SPEI (Prior Year)	−0.002 (0.002)	−0.010*** (0.003)	−0.002*** (0.001)	−0.00003 (0.001)
SPEI x Network Strength	0.020 (0.015)	0.069*** (0.023)	−0.014 (0.009)	−0.002 (0.009)
Year Fixed Effects	N	N	N	N
Producers	503	503	503	503
Years	12	23	12	23
Observations	6,036	11,569	5,533	11,066
Adjusted R ²	0.067	0.059	0.009	0.001

Robustness: SLX Coffee — Drop Year FE

Direct network and coffee spillover channels robust to dropping year FE.

	Joins Coffee Cooperative (1=Yes)					
	(1)	(2)	(3)	(4)	(5)	(6)
Direct Effect - Network	0.313*** (0.024)	0.363*** (0.023)	0.394*** (0.022)	0.375*** (0.014)	0.428*** (0.013)	0.427*** (0.013)
Direct Effect - Severe Drought	0.004 (0.009)	0.007 (0.009)	0.005 (0.008)	-0.006 (0.008)	0.006 (0.008)	-0.011 (0.007)
Direct Effect - Extreme Drought	-0.029 (0.021)	-0.020 (0.024)	0.001 (0.018)	-0.031 (0.019)	-0.008 (0.020)	-0.020 (0.015)
Indirect Effect - Network	0.671*** (0.029)	0.576*** (0.026)	0.518*** (0.023)	0.640*** (0.016)	0.616*** (0.015)	0.645*** (0.016)
Indirect Effect - Severe Drought	0.016* (0.009)	0.014 (0.009)	0.017** (0.008)	0.033*** (0.008)	0.025*** (0.008)	0.043*** (0.007)
Indirect Effect - Extreme Drought	0.018 (0.022)	0.008 (0.025)	-0.022 (0.019)	-0.006 (0.020)	-0.027 (0.021)	-0.045*** (0.017)
Weighting Formula	Binary	Inv Dist	Inv Dist	Binary	Inv Dist	Inv Dist
Other Villages	Region Only	Region Only	All	Region Only	Region Only	All
Years	12	12	12	23	23	23
Log-Likelihood	719	665	618	-403	-441	-455
Observations	6,036	6,036	6,036	11,569	11,569	11,569

Robustness: SLX Honey — Drop Year FE

The negative indirect honey effect is FE-dependent — does not survive dropping year FE.

	Joins Honey Cooperative (1=Yes)					
	(1)	(2)	(3)	(4)	(5)	(6)
Direct Effect - Network	0.521*** (0.058)	0.533*** (0.058)	0.533*** (0.059)	0.441*** (0.034)	0.452*** (0.034)	0.418*** (0.035)
Direct Effect - Severe Drought	-0.002 (0.002)	-0.004 (0.004)	-0.003 (0.004)	-0.002 (0.002)	0.005 (0.004)	0.001 (0.003)
Direct Effect - Extreme Drought	-0.003 (0.006)	-0.009 (0.012)	-0.005 (0.008)	-0.009 (0.006)	0.012 (0.009)	0.005 (0.006)
Indirect Effect - Network	0.202*** (0.053)	-0.007 (0.025)	-0.012 (0.040)	0.311*** (0.035)	0.118*** (0.026)	0.380*** (0.044)
Indirect Effect - Severe Drought	0.003 (0.003)	0.004 (0.004)	0.003 (0.005)	0.005* (0.003)	-0.001 (0.004)	0.001 (0.004)
Indirect Effect - Extreme Drought	0.005 (0.006)	0.010 (0.012)	0.006 (0.008)	0.015** (0.006)	-0.007 (0.010)	-0.003 (0.006)
Weighting Formula	Binary	Inv Dist	Inv Dist	Binary	Inv Dist	Inv Dist
Other Villages	Region Only	Region Only	All	Region Only	Region Only	All
Years	12	12	12	23	23	23
Log-Likelihood	7011	7005	7003	8553	8526	8595
Observations	6,036	6,036	6,036	11,569	11,569	11,569

Robustness: SLX Coffee — Continuous SPEI

Cross-village drought spillover for coffee persists under continuous SPEI (SPEI lower = drier).

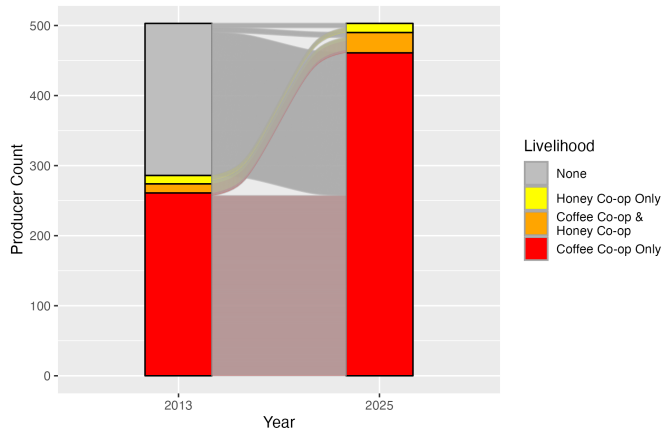
	Joins Coffee Cooperative (1=Yes)					
	(1)	(2)	(3)	(4)	(5)	(6)
Direct Effect - Network	0.312*** (0.025)	0.363*** (0.023)	0.394*** (0.022)	0.375*** (0.014)	0.428*** (0.013)	0.428*** (0.013)
Direct Effect - SPEI	-0.079 (0.061)	0.017 (0.046)	-0.025 (0.047)	0.050 (0.037)	-0.010 (0.025)	0.039 (0.027)
Indirect Effect - Network	0.675*** (0.029)	0.578*** (0.026)	0.518*** (0.023)	0.639*** (0.016)	0.614*** (0.015)	0.636*** (0.016)
Indirect Effect - SPEI	0.052 (0.061)	-0.045 (0.047)	-0.005 (0.047)	-0.100*** (0.037)	-0.048* (0.026)	-0.091*** (0.028)
Weighting Formula	Binary	Inv Dist	Inv Dist	Binary	Inv Dist	Inv Dist
Other Villages	Region Only	Region Only	All	Region Only	Region Only	All
Years	12	12	12	23	23	23
Log-Likelihood	691	634	583	-427	-466	-511
Observations	6,036	6,036	6,036	11,569	11,569	11,569

Robustness: SLX Honey — Continuous SPEI

Honey: direct network effect only; no drought spillover under continuous SPEI.

	Joins Honey Cooperative (1=Yes)					
	(1)	(2)	(3)	(4)	(5)	(6)
Direct Effect - Network	0.522*** (0.058)	0.533*** (0.058)	0.534*** (0.058)	0.442*** (0.034)	0.454*** (0.034)	0.421*** (0.035)
Direct Effect - SPEI	0.005 (0.019)	0.032 (0.026)	0.041 (0.044)	0.050*** (0.014)	−0.013 (0.017)	−0.019 (0.017)
Indirect Effect - Network	0.203*** (0.054)	−0.006 (0.025)	−0.009 (0.042)	0.324*** (0.035)	0.122*** (0.026)	0.396*** (0.046)
Indirect Effect - SPEI	−0.008 (0.019)	−0.033 (0.026)	−0.043 (0.044)	−0.055*** (0.014)	0.006 (0.017)	0.017 (0.018)
Weighting Formula	Binary	Inv Dist	Inv Dist	Binary	Inv Dist	Inv Dist
Other Villages	Region Only	Region Only	All	Region Only	Region Only	All
Years	12	12	12	23	23	23
Log-Likelihood	7011	7007	7006	8552	8519	8593
Observations	6,036	6,036	6,036	11,569	11,569	11,569

Livelihood Choices 2013 vs 2025



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